WORKED SOLUTION

STEP II 1993, Question 14

Mechanics • Energy

STEP II 1993, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q14

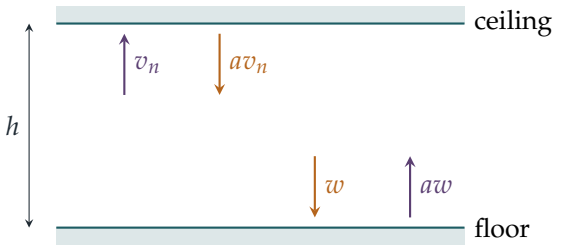
The key idea

Between impacts the ball moves freely under gravity, so $v^2 = u^2 \pm 2gh$ links its speeds at the floor and the ceiling, and each impact multiplies the speed by a. Following one full cycle (ceiling, floor, back to the ceiling) gives a linear recurrence for the kinetic energy at the ceiling, which we solve by shifting to its fixed point. The ball reaches the ceiling for the nth time exactly when that energy is positive, and taking logarithms turns this into the bound.

Solution

We model the ball as a particle moving vertically, with no air resistance, and take $0 < a < 1$ (the bound has $\ln a$ in the denominator, and if $a = 1$ the ball would hit the ceiling forever). Write v_n for the speed just before the nth ceiling impact, and

$$E_n = \frac{v_n^2}{2gh}, \qquad \text{so that} \qquad \tfrac{1}{2}mv_n^2 = mgh\,E_n.$$



The first ceiling impact

The ball leaves the floor with $u^2 = 2kgh$ and rises a height h against gravity, so

$$v_1^2 = 2kgh - 2gh = 2gh(k-1), \qquad E_1 = k - 1 > 0,$$

which is positive because $k > 1$; so the ball certainly reaches the ceiling once.

One full cycle

Just after the nth ceiling impact the ball moves down with speed av_n (Newton's experimental law at the fixed ceiling). It falls a height h, so it reaches the floor with speed w, where

$$w^2 = a^2 v_n^2 + 2gh.$$

It rebounds upwards with speed aw and rises a height h. If it gets back to the ceiling, its speed there satisfies

$$v_{n+1}^2 = a^2 w^2 - 2gh = a^4 v_n^2 + 2gh\,(a^2 - 1).$$

Dividing by $2gh$,

$$E_{n+1} = a^4 E_n + (a^2 - 1).$$

This is valid whenever the right-hand side is positive; if it is zero or negative, the ball never gets back to the ceiling.

? **Ask yourself** How do we solve a recurrence of the form $E_{n+1} = a^4 E_n + c$?

Look for the constant solution E with $E = a^4 E + (a^2 - 1)$:

$$E = \frac{a^2 - 1}{1 - a^4} = \frac{a^2 - 1}{(1 - a^2)(1 + a^2)} = -\frac{1}{1 + a^2}.$$

Subtracting this fixed point removes the constant term: putting $F_n = E_n + \dfrac{1}{1 + a^2}$ gives $F_{n+1} = a^4 F_n$. Hence $F_n = a^{4(n-1)} F_1$, and since $F_1 = (k - 1) + \dfrac{1}{1 + a^2}$,

$$E_n = a^{4n-4}\left(k - 1 + \frac{1}{a^2 + 1}\right) - \frac{1}{a^2 + 1} = a^{4n-4}(k - 1) + \frac{a^{4n-4} - 1}{a^2 + 1}.$$

Multiplying by mgh, the kinetic energy just before the nth ceiling impact is

$$mgh\left(a^{4n-4}(k - 1) + \frac{a^{4n-4} - 1}{a^2 + 1}\right),$$

as required.

The number of ceiling impacts

The ball hits the ceiling for the nth time only if it has a positive speed on arriving there, that is, only if $E_n > 0$. (If it hits the ceiling an nth time it has hit it on every earlier occasion too, so the formula above applies.) Multiplying $E_n > 0$ by $a^2 + 1 > 0$,

$$a^{4n-4}((k - 1)(a^2 + 1) + 1) > 1 \quad \Longleftrightarrow \quad a^{4n-4}(a^2(k - 1) + k) > 1.$$

Both sides are positive, so we may take logarithms:

$$(4n - 4)\ln a + \ln[a^2(k - 1) + k] > 0.$$

Ask yourself Which way does the inequality go when we divide by $4\ln a$?

Since $0 < a < 1$, we have $\ln a < 0$, so dividing by $4\ln a$ reverses the inequality:

$$n - 1 < -\frac{\ln[a^2(k-1)+k]}{4\ln a}, \qquad \text{that is,} \qquad n < 1 - \frac{\ln[a^2(k-1)+k]}{4\ln a}.$$

So if the ball hits the ceiling N times, then the inequality holds with $n = N$ and

$$N < 1 - \frac{\ln[a^2(k-1)+k]}{4\ln a}.$$

In particular the number of ceiling impacts is at most this value, which is what we had to show. (In fact E_n decreases with n, so N is exactly the number of positive integers strictly below the bound.)

Checks

- $n = 1$: the formula gives $mgh(k-1)$, matching $\frac{1}{2}m(2kgh) - mgh$ from energy conservation.
- Large n: $E_n \to -\frac{1}{1+a^2} < 0$, so the ball does eventually stop reaching the ceiling, as it must when energy is lost at every impact.
- Numerically, with $k = 6$, $a = 0.9$, $h = 2\,\text{m}$, $m = 1\,\text{kg}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, a step-by-step simulation gives ceiling kinetic energies 98, 60.57, 36.02 J, agreeing with the formula. The ball hits the ceiling 6 times, and the bound is about 6.48.

Common slips

- Forgetting that the speed is multiplied by a at *both* the ceiling and the floor, which gives a^2 instead of a^4 per cycle.
- Dividing by $\ln a$ without reversing the inequality.
- Applying restitution to the kinetic energy rather than to the speed: the energy is multiplied by a^2 at each impact, not by a.

Beyond STEP

The recurrence $E_{n+1} = a^4 E_n + c$ is a first-order linear difference equation. The trick of subtracting the fixed point is the discrete version of finding a particular integral for a linear ODE, and it reappears in the study of iterated maps and their stability. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and energy loss more generally.

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